Table 4. Regime-conditioned within-family selection used after Definition 4.1 chooses a watermark family. For each estimated edit regime, the concrete scheme is the method with the highest regime-conditioned score $S_r(m)$.

Family	**Estimated edit regime**	**Regime-conditioned score $S_r(m)$ with tuple (AUC, z)**	**Selected representative**	**Watermarking Parameters**
Token-level	$\hat{\varepsilon} \approx 0.15$	**HCW: 1.168** $(0.920, 3.03)$ DiPMark: 1.129 $(0.907, 3.50)$ HeavyWater: 1.103 $(0.887, 3.63)$ SimplexWater: 1.084 $(0.877, 3.83)$ Unigram: 0.904 $(0.790, 7.77)$ KGW: 0.888 $(0.780, 8.27)$	**HCW**	δ-reweight variant (`method=delta`)
	$\hat{\varepsilon} \approx 0.25$	**HCW: 1.137** $(0.910, 3.40)$ DiPMark: 1.104 $(0.900, 3.90)$ HeavyWater: 1.072 $(0.880, 4.20)$ SimplexWater: 1.052 $(0.870, 4.50)$ Unigram: 0.982 $(0.880, 8.80)$ KGW: 0.954 $(0.860, 9.60)$	**HCW**	δ-reweight variant (`method=delta`)
Semantic	$\hat{\varepsilon} \approx 0.42$ $\hat{\varepsilon}_s \approx 0.10$	**PMark: 1.305** $(0.850, 1.20)$ SemStamp: 1.220 $(0.820, 1.50)$ SimMark: 1.170 $(0.800, 1.70)$	**PMark**	Rejection parameter $\rho = 0.25$
Distribution-preserving	$\hat{\varepsilon} \approx 0$	**CGW: 1.990** $(0.990, -5.80)$	**CGW**	Security parameter $\lambda = 128$, fixed secret key across runs

Note. This table explains the second stage of the Hybrid. Definition 4.1 first selects the family from the estimated edit regime. The concrete scheme is then chosen within that family using the regime-conditioned score $S_r(m) = \mathrm{AUC}_r(m) + \frac{1}{1+\max(z_r(m),0)}$. For the token-level row at $\hat{\varepsilon} \approx 0.15$, the score is averaged over the attacks in Table 2 with that edit rate, namely OPT-2.7B paraphrasing, watermark-removal prompting, and synonym substitution. The token-level row at $\hat{\varepsilon} \approx 0.25$ uses the DIPPER regime. The semantic row uses the back-translation regime, for which $(\hat{\varepsilon}, \hat{\varepsilon}_s) \approx (0.42, 0.10)$. Greener cells indicate higher within-family scores and therefore the selected representative at that regime.